REPUBLIC OF UGANDA

Ministry of Agriculture, Animal Industry and Fisheries

DATABASE PROGRAMMING PROJECT

Agricultural Services Database for Coffee Growers Management

Submitted to:

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1. Project Overview

The Ministry of Agriculture, Animal Industry and Fisheries (MAAIF) of Uganda requires a centralised, secure relational database to support its Coffee Growers Management Programme. The programme aims to register all coffee farmers, track farm-level production, manage distribution of agricultural inputs, and facilitate effective communication between farmers and extension workers.

This document presents the design, implementation, security features, and testing outcomes of the Agricultural Services Database developed using MySQL and VS Code. The system addresses the full data lifecycle: from farmer registration and farm profiling to input distribution, farm visits, and regional performance reporting.

2. Requirements Specification

2.1 Functional Requirements

- Uniquely register coffee growers using a farmerID despite their National Identification Number (NIN).
- Maintain a complete profile for each farmer including personal details, farm location, and contact information.
- Record and track harvest data (bags of coffee) per farm each season.
- Calculate estimated yield based on registered number of trees per farm.
- Track quantity and type of agricultural inputs (seedlings, fertiliser, etc.) distributed to each farmer.
- Manage an inventory of available support items and update stock after each distribution event.
- Prevent double-dipping: a farm cannot receive the same support type more than once per distribution session.
- Allow extension workers to log farm visit reports including date, advice given, and pests or diseases possibly observed.
- Provide a visit-request feature allowing farmers to flag a need for technical assistance.
- Generate reports showing total coffee production per region, identifying over-performing and under-performing regions.
- Allow grouping of farmers by Region, District, and Sub-county for localised planning and support.
- Maintain a directory of extension workers and their assigned regions.
- Enforce data privacy and access control through user roles and privileges.

2.2 Non-Functional Requirements

- Data integrity (and referential integrity) enforced through primary keys, foreign keys, and check constraints.
- System must comply with the Uganda Data Protection and Privacy Act (truncated emails i.e. “mug***@***”).

- Only authorised administrators may modify inventory records.
- The system must support concurrent access by multiple users with role-based restrictions.
- Database must support backup and recovery to protect against data loss.

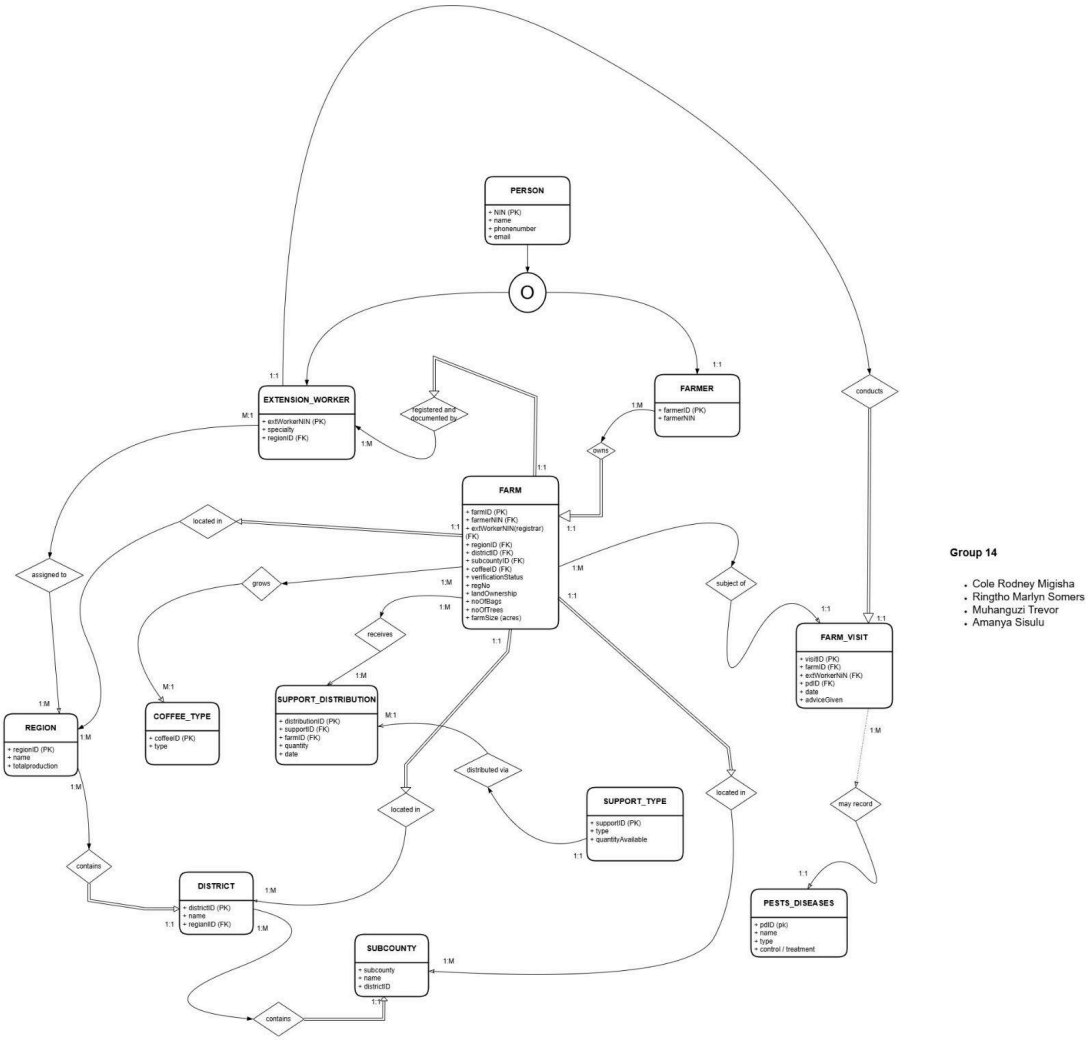
3. System Assumptions

- Every coffee farmer is assumed to possess a valid Uganda National Identification Number (NIN).
- Every farm grows only one type of coffee for the purposes of this system.
- Land ownership is exclusive: one plot of land belongs to exactly one farmer.
- All data entered into the system is assumed to be pre-verified by the responsible officer.
- Extension workers are responsible for updating the central database after each field visit or data collection exercise.
- Every farmer must have at least one phone number on record; an email address is optional.
- The system is deployed within a trusted institutional network and is not publicly accessible.
- All extension workers are assigned to a specific region.

4. Database Design

4.1 Enhanced Entity-Relationship Diagram (EERD)

The EERD below illustrates the full logical model of the database. It includes a generalisation hierarchy (PERSON is the supertype with FARMER and EXTENSION_WORKER as overlap subtypes), all entity sets, relationships with their cardinalities, and key business rules.



Rules

- A PERSON must be either a FARMER or an EXTENSION_WORKER or both (overlap specialisation).
- A FARMER must own at least one FARM; every FARM is owned by exactly one FARMER.
- An EXTENSION_WORKER must be assigned to exactly one REGION.
- An EXTENSION_WORKER registers and documents at least one FARM; every FARM has exactly one registrar.
- Every FARM must be located in exactly one REGION, one DISTRICT, and one SUB-COUNTY.
- A FARM may receive zero or many SUPPORT_DISTRIBUTION events (1:M); each event links to one FARM.
- A SUPPORT_TYPE can be distributed via many SUPPORT_DISTRIBUTION events (M:1).
- Each FARM_VISIT is conducted by exactly one EXTENSION_WORKER and linked to exactly one FARM.
- A FARM_VISIT can record even no PESTS_DISEASES occurrence.
- A REGION contains one or many DISTRICTs; a DISTRICT belongs to exactly one REGION (1:M).
- A DISTRICT contains one or many SUB-COUNTYs; each SUB-COUNTY belongs to exactly one DISTRICT (1:M).

Figure 1: Enhanced Entity-Relationship Diagram — Agricultural Services Database

4.2 Business Rules

- A PERSON must be either a FARMER or an EXTENSION_WORKER or both (overlap specialisation).
- A FARMER must own at least one FARM; every FARM is owned by exactly one FARMER.
- An EXTENSION_WORKER must be assigned to exactly one REGION.
- An EXTENSION_WORKER registers and documents at least one FARM; every FARM has exactly one registrar.
- Every FARM must be located in exactly one REGION, one DISTRICT, and one SUB-COUNTY.
- A FARM may receive zero or many SUPPORT_DISTRIBUTION events (1:M); each event links to one FARM.
- A SUPPORT_TYPE can be distributed via many SUPPORT_DISTRIBUTION events (M:1).
- Each FARM_VISIT is conducted by exactly one EXTENSION_WORKER and linked to exactly one FARM.
- A FARM_VISIT can record even no PESTS_DISEASES occurrence.
- A REGION contains one or many DISTRICTs; a DISTRICT belongs to exactly one REGION (1:M).
- A DISTRICT contains one or many SUB-COUNTYs; each SUB-COUNTY belongs to exactly one DISTRICT (1:M).

4.3 Table Summary

Table	Primary Key	Description
Person	NIN	Supertype: all individuals in the system
Farmer	FarmerID	Subtype: registered coffee growers
ExtensionWorker	ExtWorkerNIN	Subtype: ministry field officers
Farm	FarmID	Central entity: farm details & location
Region	RegionID	Geographic level 1
District	DistrictID	Geographic level 2
SubCounty	SubCountyID	Geographic level 3
CoffeeTypes	CoffeeID	Arabica, Robusta, Liberica, etc. (too many to use enum)
SupportTypes	SupportID	Inventory of available inputs
SupportDistribution	DistributionID	Input distribution events per farm
FarmVisit	VisitID	Extension worker visit logs
VisitRequest	RequestID	Farmer-initiated assistance requests

PestDisease	PdID	Known pests and diseases with controls

5. Database Implementation

5.1 Tools & Technologies

Component	Details
DBMS	MySQL 8.0.45.0
IDE / Editor	Visual Studio Code with MySQL extension
Language	SQL
OS	Windows

5.2 Data Validation Constraints

All tables are designed with the following validation strategy:

- PRIMARY KEY constraints on every table to enforce entity uniqueness.
- FOREIGN KEY constraints with ON UPDATE CASCADE to maintain referential integrity across all relationships.
- UNIQUE constraints: Farm RegNo, Person NIN, SupportTypes.Type, and the composite (FarmID, SupportID, DistributionDate) in SupportDistribution to prevent double-dipping.
- CHECK constraints: FarmSizeAcres > 0, NoOfTrees >= 0, NoOfBags >= 0, QuantityAvailable >= 0.
- ENUM constraints on VerificationStatus ('Pending','Verified','Rejected'), LandOwnership ('Owned','Leased','Family','Communal'), PestDisease.Type ('Pest','Disease'), and VisitRequest.Status (‘Pending’, ‘Assigned’, ‘Completed’, ‘Cancelled’).
- NOT NULL constraints on all mandatory fields; nullable Email field honours the optional-contact assumption.
- DEFAULT values: VerificationStatus defaults to 'Pending'; QuantityAvailable and NoOfBags default to 0.

5.3 Key SQL Structures

Farm Table (central entity):

Column	Data Type	Constraint / Note
FarmID	INT	PRIMARY KEY, AUTO_INCREMENT
VerificationStatus	ENUM	Pending | Verified | Rejected; DEFAULT Pending
RegNo	VARCHAR(50)	UNIQUE — ministry registration number
NoOfBags	INT	CHECK >= 0; updated by extension workers
FarmSizeAcres	DECIMAL(8,2)	CHECK > 0
NoOfTrees	INT	CHECK >= 0; used for yield estimation
CoffeeID	INT (FK)	References CoffeeTypes
ExtWorkerNIN	VARCHAR(14) FK	References Extension Worker (registrar)

6. Security & Automation

6.1 User Roles & Privileges

Four user roles are defined to enforce the principle of least privilege:

User Role	Username	Permissions
Administrator	admin_user	FULL access (ALL PRIVILEGES) — all tables, views, procedures, inventory management
Extension Worker	ext_worker_user	Log visits, update farms, view farmer directory, view requests
Report Compiler	report_user	Read-only access to all views and production report stored procedure
Farmer	farmer_user	View own farm data, distribution history, raise visit requests

6.2 Views

Seven views are implemented to restrict direct table access and simplify reporting:

- **vw_production_by_region** — Provides farm count, tree count, bags harvested, and estimated yield per region.
- **vw_region_performance** — Classifies each region as Over-Performing, Under-Performing, or On-Target compared to the national average.

- **vw_inventory_status** — Shows current stock, total distributed, and original stock per support type.
- **vw_farmer_directory** — Displays farmer profiles with masked email addresses for UDPPA compliance.
- **vw_pending_requests** — Lists all pending visit requests with farmer contact details and region.
- **vw_distribution_history** — Tracks full distribution history linking farms, farmers, and support types.
- **vw_ext_worker_directory** — Displays extension workers along with their assigned regions.

6.3 Stored Procedures

- **sp_register_farmer** — Atomically inserts a new farmer record with validation for invalid registration cases.
- **sp_register_ext_worker** — Registers a new extension worker.
- **sp_register_dual_role** — Registers a person in both farmer and extension worker roles.
- **sp_log_farm_visit** — Logs a farm visit report.
- **sp_distribute_support** — Validates available stock before distributing support and updating inventory.
- **sp_production_report** — Returns a regional performance report.
- **sp_farms_by_location** — Filters farms by region, district, or sub-county.

6.4 Triggers

- **trg_farmer_role_check —** Enforces role overlap using the isFarmer flag.
- **trg_ew_role_check —** Enforces role overlap using the isExtWorker flag.
- **trg_inventory_check —** Blocks distribution if stock is insufficient.
- **trg_no_double_dip —** Ensures farms only receive the designated amount per month.
- **trg_update_inventory —** Automatically deducts stock after distribution.
- **trg_update_region_production —** Recalculates regional totals when bag counts are updated.
- **trg_verified_farm_only —** Ensures support only reaches ministry-approved (verified) farms

7. Backup & Recovery Strategy

7.1 Backup Approach

We implemented a full logical backup using mysqldump. The dump file contains all DDL (CREATE statements) and DML (INSERT statements), enabling complete database reconstruction on any MySQL server.

7.2 Backup Command

Full backup command:

```
mysqldump -u root -p agri_services_db > ("agri_backup_" + (Get-Date -Format "yyyyMMdd") + ".sql")
```

Created the dump file — *agri_backup_20260403.sql*

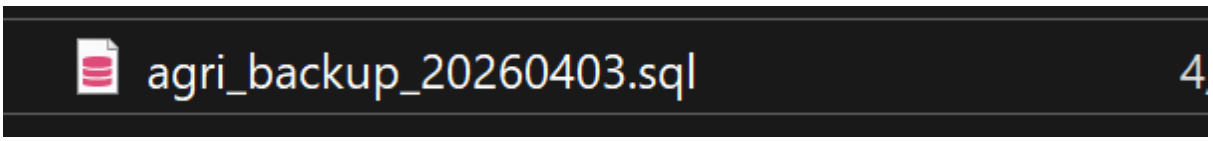


Figure 2: Screenshot showing the sql dump created

Restore command:

```
mysql -u root -p agri_coffee_db < agri_backup_20260403.sql
```

8. Testing

8.1 Test Cases

#	Test Case	Expected Result	Actual Result	Status
1	Register a farmer with valid NIN (TEST-NIN-001) and farmerID (FMR-TEST-001) via sp_register_farmer	Farmer and Person records inserted successfully	As expected	**PASS**
2	Register duplicate NIN (TEST-NIN-001) via sp_register_farmer	Error: Duplicate entry for PRIMARY KEY on PERSON.NIN	As expected	**PASS**
3	Insert farm with farmSizeAcres = 0.00 via direct INSERT	Error: CHECK constraint violation — farmSizeAcres > 0	As expected	**PASS**

4	Distribute support to unverified farm (FM004, verificationStatus = PENDING) via sp_distribute_support after adding trg_verified_farm_only	Trigger fires: 'Support can only be distributed to verified farms.'	As expected	**PASS**
5	Distribute same support type (S01) to farm FM001 twice in the same month — first call 400 units, second call 200 units (total 600, exceeds quota of 500)	Second call blocked by trg_no_double_dip: 'Double-dip violation: monthly quota exceeded for this farm and support type.'	As expected	**PASS**
6	Update noOfBags on farm FM001 (Region R01) to 999 via UPDATE FARM	trg_update_region_production fires automatically — REGION.totalProduction for R01 recalculated to reflect new value	As expected	**PASS**
7	Call sp_production_report( )	Returns all regions with OVER-PERFORMING / UNDER-PERFORMING / ON-TARGET classification against national average	As expected	**PASS**
8	Log farm visit for FM002 with pest PD01 (Coffee Berry Borer) via sp_log_farm_visit	FARM_VISIT record inserted with pdID = PD01 linked to PESTS_DISEASES; advice text stored	As expected	**PASS**
9	Connect as report_user and run SELECT * FROM vw_farmer_directory	Returns all farmer profiles with email addresses masked (e.g. mug***@***) — SELECT succeeds	As expected	**PASS**
10	Connect as report_user and attempt INSERT into FARM_VISIT	Error: INSERT command denied to user 'report_user'@'localhost' for table 'FARM_VISIT' (ERROR 1142)	As expected	**PASS**

```
mysql> select * from vw_farmer_directory
    -> ;
+----------+------------+------------------+-------------+-------------+--------+----------+------------+-------------+--------------------+
| farmerID | farmerNIN  | pe_name          | phoneNumber | maskedEmail | farmID | regionID | districtID | subcountyID | verificationStatus |
+----------+------------+------------------+-------------+-------------+--------+----------+------------+-------------+--------------------+
| FMR-001  | CF-NIN-001 | Mugisha John     | 0704100001  | mug***@***  | FM001  | R01      | D01        | SC01        | VERIFIED           |
| FMR-002  | CF-NIN-002 | Apio Grace       | 0714100002  | NULL        | FM002  | R01      | D02        | SC03        | VERIFIED           |
| FMR-003  | CF-NIN-003 | Ssekandi Peter   | 0724100003  | sse***@***  | FM003  | R02      | D04        | SC04        | VERIFIED           |
| FMR-004  | CF-NIN-004 | Namukasa Annet   | 0734100004  | NULL        | FM004  | R01      | D01        | SC02        | PENDING            |
| FMR-005  | CF-NIN-005 | Byamukama Daniel | 0744100005  | bya***@***  | FM005  | R03      | D05        | SC05        | VERIFIED           |
| FMR-001  | CF-NIN-001 | Mugisha John     | 0704100001  | mug***@***  | FM010  | R04      | D010       | SC010       | VERIFIED           |
+----------+------------+------------------+-------------+-------------+--------+----------+------------+-------------+--------------------+
6 rows in set (0.02 sec)
```

Figure 3: Results in terminal from tests 9 and 10 after logging into report_user

```
mysql> INSERT INTO FARM_VISIT (visitID, farmID, extWorkerNIN, visitDate, adviceGiven)VALUES ('FV-HACK', 'FM001', 'EW-NIN-001', CURDATE(), 'Unauthorised entry attempt');
ERROR 1142 (42000): INSERT command denied to user 'report_user'@'localhost' for table 'farm_visit'
mysql>
```

9. Conclusion

The Agricultural Services Database successfully fulfils all requirements outlined in the Ministry of Agriculture's Coffee Growers Management Programme brief. The system provides a robust, secure, and scalable foundation for registering and tracking coffee farmers across Uganda's regions.

The system is ready for deployment and further enhancement, including a future web-based front-end interface to improve field usability for extension workers and report making officers.